Reproducing the MOVES6 on-road emission-rate engine in R, row by row, for use in vehicular emission inventories of the Washington DC area

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Abstract

The U.S. EPA Motor Vehicle Emission Simulator (MOVES) is the regulatory model for on-road emissions in the United States, but running it requires a Java application, a MariaDB server, compiled Go programs and a full database installation. This makes MOVES hard to embed in inventory models such as VEIN. We describe movesr, an R package that reimplements the MOVES6.0.0 rate engine using the MOVES default database stored as Parquet files. The port covers running and start exhaust, brake and tire wear, evaporative permeation, vapor venting and fuel leaks (parked and while driving), and the chained pollutants (CH4, NMHC, NMOG, TOG, VOC, CO2, CO2e, SO2, PM2.5 species and PM10). Each MOVES step was translated from the Java, SQL and Go source code and checked against MOVES execution tables and final outputs. We compared 4,302,126 MOVES6 rates from five reference runs (Washtenaw County MI and the District of Columbia, Montgomery County MD and Fairfax County VA). Every rate was reproduced. For exhaust, brake and tire rates the largest relative difference was 1.6×10^{-5}, which is at the level of the six significant digits that MOVES writes. Evaporative rates agreed within 2.9×10^{-5} at the 99th percentile; the largest difference, 8.7×10^{-4}, occurred in E85 vapor venting, which amplifies single-precision rounding of tank temperatures. We also describe a simplified evaporative method that replaces the MOVES trip simulation by interpolation in a library of exact results. It reproduces permeation and leak daily totals within 0.44% and vapor venting within 3.8% on average, but saves little time in a single run. On the same workstation, movesr produced the rates of each reference run 3.7 to 9.2 times faster than MOVES with one worker, using about three times more memory than the largest MOVES process. Factors written by movesr are read by VEIN without loss. The paper documents the procedure that makes the numbers identical across MOVES, movesr and VEIN, and the MOVES behaviours that a reimplementation must reproduce.

Keywords: MOVES, Emission factors, Vehicular emissions inventory, Evaporative emissions, Reproducibility, R

1. Introduction

Bottom-up vehicular emission inventories combine traffic activity with emission factors. In the United States the regulatory source of on-road emission factors is the Motor Vehicle Emission Simulator, MOVES (U.S. Environmental Protection Agency, 2026a, 2026b). MOVES combines a large default database (fleet, activity, fuels, meteorology, inspection and maintenance programs) with emission rates by operating mode, and produces either inventories or rate tables. EPA guidance allows MOVES to be run in "rates" mode so that its output, in grams per mile, per start or per vehicle, can be multiplied by activity in an external model (U.S. Environmental Protection Agency, 2026b). The structure and content of the MOVES input data are described by Koupal et al. (2014).

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Several groups have reduced the cost of using MOVES in large applications. Wu et al. (2014) and Liu (2015) built rate databases from MOVES runs for site-specific and regional analyses. MOVES-Matrix runs MOVES over grids of inputs on distributed computers and stores the resulting rates for lookup (Liu et al., 2019, 2017), and was later extended to engine starts, soak, evaporative and truck hotelling emissions (Xu et al., 2018). These approaches still need MOVES to produce every rate; a combination of inputs outside the stored grid requires new MOVES runs.

VEIN is an R package for vehicular emission inventories (Ibarra-Espinosa, 2026a; Ibarra-Espinosa et al., 2018b), and eixport passes its emissions to atmospheric models (Ibarra-Espinosa et al., 2018a). VEIN has been used, for example, to model ozone and NOx in São Paulo street canyons (Gavidia-Calderón et al., 2021). Related work by the author covers tunnel measurements of emission factors in São Paulo (Nogueira et al., 2021), a century of vehicular emissions in Brazil and their effect on air quality (Ibarra-Espinosa et al., 2026), the effect of evaporative emission controls on ozone (Ibarra-Espinosa et al., 2022), high emitters in Ecuador's inspection and maintenance program (Ibarra-Espinosa et al., 2025), the spatial disaggregation of vehicular inventories (Meotti et al., 2025), and WRF-Chem simulations of Brazilian cities (Schuch et al., 2026).

VEIN already reads MOVES rate tables, but producing those tables still requires MOVES itself. MOVES6 is a Java application that stores every intermediate table in MariaDB, delegates parts of the computation to programs written in Go, and needs the complete default database to be installed. A single run for three counties, two hours and all source types took close to an hour on our workstation with one worker (Section 3.6). For an inventory with many counties, months and hours this is a practical limit, and the dependency stack makes it hard to run MOVES inside other modelling systems or to inspect intermediate results.

This paper describes movesr (Ibarra-Espinosa, 2026b), an R package that reproduces the MOVES6.0.0 rate engine without Java, MariaDB or MOVES. The objective is narrow: for the same inputs, movesr must return the same numbers as MOVES, rate by rate, so that an inventory built with VEIN gives the same result whether the factors came from MOVES or from movesr. We do not propose new emission science. We document how each MOVES step was reproduced, how equality was verified, which MOVES behaviours a faithful port must keep, and where movesr departs from MOVES on purpose (a simplified evaporative method). The work prepares a VEIN inventory for the Washington DC area, so the reference runs use the District of Columbia and two neighbouring counties.

The code was written with the help of a large language model working as a coding agent. Section 6 describes how, and which model. We report this because it matters for how much trust readers should place in the software: the evidence of correctness in this paper comes from comparison with MOVES outputs, not from the way the code was written.

2. Methods

2.1. Reference model and reference runs

The reference is MOVES6.0.0 (U.S. Environmental Protection Agency, 2026c), source commit `45805553` (full hash in the reference list), with the default database `movesdb20260729`. MOVES was compiled and run on Linux with Java 11.0.10, Go 1.26.10 and MariaDB 10.11.14, using a private database server and a single worker, so that runs were deterministic and isolated from other work. Five reference runs ("golden sets") were made with the real model (Table 1). The RunSpecs are stored in the movesr repository (`data-raw/dmv/*.mrs`) and in VEIN (`inst/extdata/moves/*.mrs`), together with the script that starts the database, runs MOVES, exports the output tables and shuts the server down (`data-raw/dmv/generate_dmv.py`). Besides the final rate tables, the script exports MOVES execution tables (activity, tank temperatures, soak fractions, coefficients) so that intermediate steps can be checked one at a time.

The two Washtenaw runs are the reference data distributed with VEIN. The three DMV counties belong to the Washington DC-MD-VA ozone nonattainment area (U.S. Environmental Protection Agency, 2026d); movesr includes the 15 jurisdictions of that area as a predefined region.

2.2. Porting rules

The port follows four rules.

Table 1: MOVES6 reference runs. All runs use all 13 source types and every valid fuel, model years 1985 to 2025 (Washtenaw: 2000 to 2025), urban unrestricted roads for distance rates, and MOVES defaults for fleet, fuels and inspection programs.

Run	Location	Time	Processes
Washtenaw running	Washtenaw County MI (26161)	2025, January, weekend day, hours 1 to 3 (20, 10 and 30 C, 50% RH; 20 C in other hours)	Running (1): CO, NOx
Washtenaw events	Washtenaw County MI (26161)	Same as Washtenaw running	Start (2): CO, NOx; evaporative (11 to 13) off network: THC
DMV running	DC (11001), Montgomery MD (24031), Fairfax VA (51059)	2025, July, weekday, hours 16 and 17, default weather	Running (1), brake (9), tire (10), evaporative (11 to 13) on roads; 21 pollutants
DMV events	Same three counties	2025, July, weekday, hours 16 and 17, default weather	Start (2), evaporative (11 to 13) off network
DC 24 h evaporative	District of Columbia (11001)	2025, July, weekday, hours 1 to 24, default weather	Evaporative (11 to 13); THC, NMHC, NMOG, TOG, VOC

1. *Read the code that actually runs.* MOVES6 contains several implementations of the same step. For example, drive-cycle operating modes are computed by a Go program (`generators/baserategenerator`) because a compilation flag (`USE_EXTERNAL_GENERATOR_FOR_DRIVE_CYCLES`) is set; the Java SQL in `RatesOperatingModeDistributionGenerator` is not used and gives different results for heavy-duty cycles. Vapor venting uses `MultidayTankVaporVentingCalculator.sql` because `USE_MULTIDAY_DIURNALS` is true, not the older `TankVaporVentingCalculator.sql`. Each movesr function names the MOVES file it reproduces.
2. *Check every stage against MOVES.* When MOVES writes an intermediate table to its execution database, the corresponding movesr table was compared with it before the next stage was written.
3. *Compare all rows.* Every rate of every reference run is compared, matched on all key columns (county or zone, hour, source type, fuel type, model year, pollutant, process, road type, speed bin). A rate present in MOVES and absent in movesr, or the reverse, is reported as a failure, not dropped.
4. *Keep MOVES arithmetic.* MOVES stores most intermediate values in single-precision `FLOAT` columns and the tank temperature generator uses Java `float`. Where this changes results beyond the output precision, movesr rounds to single precision at the same points (*IEEE standard for floating-point arithmetic*, 2019). Elsewhere it uses double precision.

movesr is written in R (R Core Team, 2021) with data.table (Barrett et al., 2026). It runs in one thread and has no compiled code. Figure 1 shows how the MOVES stages map to movesr functions and where the comparison with MOVES enters.

2.3. Default database

The MOVES default database (240 tables) was converted once to Apache Parquet (The Apache Software Foundation, 2026) with DuckDB (Raasveldt and Mühleisen, 2019) through its MySQL extension. Row counts of all tables were compared with the MariaDB source (no differences), and a SHA-256 hash of each file is stored in a catalog. The Parquet files total 10 MiB. movesr reads them through the DuckDB command-line program, so it needs neither a database server nor the `arrow` R package. Tables that MOVES fills during a run (source-bin distributions, operating-mode distributions, fuel adjustments, activity) are empty in the default database; movesr computes them.

2.4. Running exhaust (process 1)

Source bins. A MOVES source bin combines fuel, engine technology, regulatory class and model-year group. Its activity share within a source type and model year is the sum of `stmyFraction` in

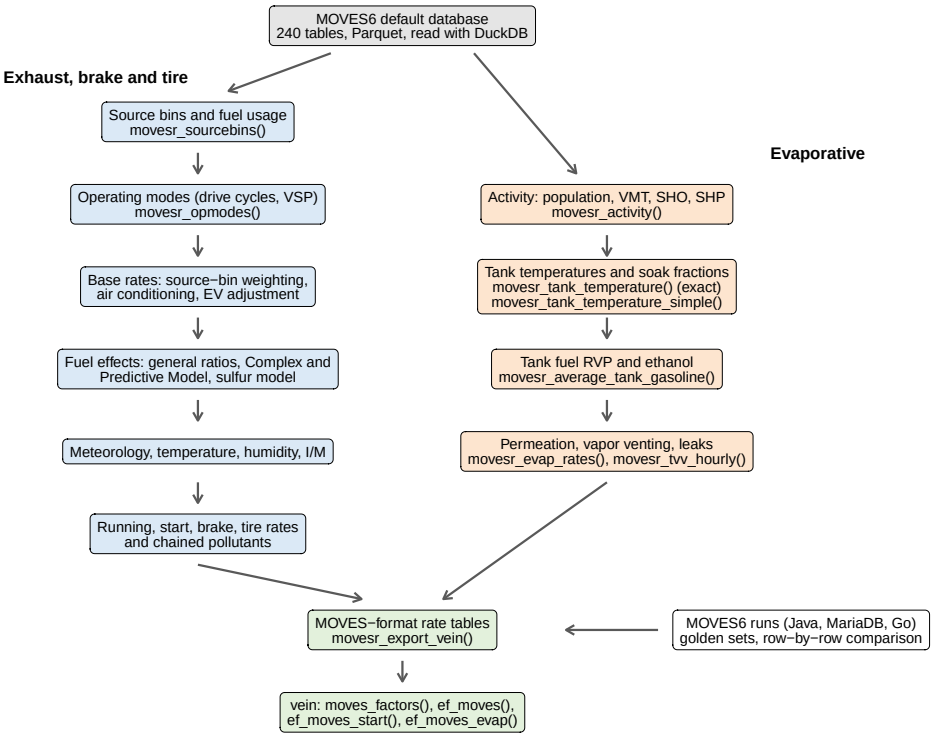


Figure 1: Structure of movesr. Each box is a MOVES stage and the movesr function that reproduces it; the reference runs of MOVES6 are compared with the rate tables that movesr writes.

SampleVehiclePopulation. Regulatory class and model-year group enter the bin only when SourceTypePolProcess requires them (isRegClassReqd, isMYGroupReqd). FuelUsageFraction then moves activity from the equipped fuel to the fuel actually used, so that, for example, most flex-fuel activity is reported as gasoline.

Operating modes. For each source type, road type and average-speed bin, MOVES brackets the bin speed between two drive schedules and weights them by distance in speed. For each second of a schedule the Go generator computes the vehicle specific power (Jiménez-Palacios, 1999) from the road-load coefficients of SourceUseTypePhysics:

$$\mathrm{VSP} = \frac{Av + Bv^2 + Cv^3 + Mva}{f_m}, \tag{1}$$

with v and a in m s^{-1} and m s^{-2} and f_m the fixed mass factor. Speed below 1 mph is idle (mode 1); braking (mode 0) is $a \le -2$ mph s^{-1}, or three consecutive seconds below -1 mph s^{-1}. The Go code sets the acceleration to zero when the previous second is missing, which happens in heavy-duty schedules; the first port computed accelerations across those gaps and failed the comparison until this was reproduced. Speed bounds of the operating modes are in mph. One distribution is produced per row of SourceUseTypePhysics (source type, regulatory class, model-year range).

Base rates. Rates are averaged over source bins with their activity shares, separately for the mean rate, the I/M rate and the air-conditioning increment:

$$\mathrm{MBR} = \frac{\sum_b s_b\, r_b}{\sum_b s_b}, \qquad \mathrm{ER} = \frac{\sum_m \mathrm{MBR}_m\, f_m}{\bar{v}} \ \ [\mathrm{g\,mile^{-1}}], \tag{2}$$

where s_b is the source-bin share, r_b the EmissionRateByAge value, f_m the operating-mode fraction and $\bar{v}$ the bin speed. An electric-vehicle adjustment raises non-electric rates when the fleet-average standard is met partly by electric sales.

Corrections. The calculator applies, per fuel formulation, the fuel-effect ratio, a temperature factor $1 + (t-75)(A + (t-75)B)$, the NOx humidity factor $K = 1 - 0.0329\,(H - 10.71)$ with H in g kg^{-1} bounded to $[3, 17.71]$, the air-conditioning factor from the heat index, and the I/M fraction $f = \sum \mathrm{IMFactor} \times \mathrm{compliance}/100$:

$$r = R \cdot T \cdot \left[(1-f)(\mathrm{ER} + \mathrm{AC}\,\mathrm{ER}_{AC}) + f(\mathrm{ER}_{IM} + \mathrm{AC}\,\mathrm{ER}_{IM,AC})\right]. \tag{3}$$

Fuel-effect ratios come from GeneralFuelRatioExpression, which stores MySQL expressions; movesr translates them to R and evaluates them for each formulation of the fuel supply. High-ethanol fuels take E10 properties as MOVES does. Model years without an expression use the Complex Model (CO) or the Predictive Model (THC, NOx) followed by the sulfur model. The result is weighted by market share and by the gasoline program area fraction.

Regulatory classes. Rates computed by regulatory class are combined with weights equal to the source-bin activity shares, which is what the Go code does in calculateActivityWeight. Ages with no population give a zero rate, as in MOVES.

2.5. Start exhaust (process 2)

Start rates use operating modes 101 to 108 (soak-time classes) from StartsOpModeDistribution, by source type, hour and day type. Each mode rate receives the start temperature adjustment (polynomial or logarithmic by fuel, pollutant and model-year group, using $\min(t, 75) - 75$), then the same fuel, temperature, I/M, electric-vehicle and regulatory-class stages as running rates, without air conditioning.

2.6. Brake and tire wear (processes 9 and 10)

Brake-wear operating modes come from the same Go drive-cycle generator as running exhaust, and tire-wear modes from the speed classes of AverageSpeedOperatingModeDistributionGenerator. Their bins carry model-year group 0, so SourceTypePolProcess decides the bin structure. PM10 comes from PM2.5 through PM10PM25Ratio.

2.7. Chained pollutants

Chained pollutants are computed per regulatory class, before aggregation. CH4 and NMHC follow from THC and `MethaneTHCRatio`; NMOG and VOC from NMHC and `HCSpeciation`, using the oxygenate content of each formulation; TOG is NMOG plus CH4. For E85 vehicles from model year 2001, NMOG and VOC use an alternative THC evaluated with E10 RVP, as in the Go calculator. CO2 follows from energy, carbon content and oxidation fraction, CO2e from CO2, CH4 and N2O with global warming potentials, and SO2 from energy and fuel sulfur. PM2.5 is split into elemental carbon, sulfate, aerosol water and the remainder, with the sulfur adjustment and the crankcase split of `SulfatePMCalculator.sql`.

2.8. Evaporative emissions (processes 11, 12, 13)

Evaporative emissions needed four parts of MOVES that the exhaust processes do not use.

Activity. From `TotalActivityGenerator` and `ActivityCalculator.sql`, movesr computes population by source type and age, travel fractions within each HPMS vehicle class, annual and hourly VMT, source hours operating (SHO = VMT / average speed) and source hours parked:

$$\mathrm{SHP} = P \cdot N_d - \sum_{\text{roads}} \mathrm{SHO}, \tag{4}$$

where P is population and N_d the number of real days in the day type (5 for weekdays). The zone population is P times the zone's share of source hours operating (`SHOAllocFactor`), averaged over road types with their VMT fractions. MOVES reports evaporative rates as emissions divided by population, so the per-vehicle rate equals the rate per parked hour times $\mathrm{SHP} \cdot \alpha/(P_z N_d)$, with α the zone allocation factor.

Tank temperatures. `TankTemperatureGenerator` simulates the sample vehicles of the day type (about 11,000 vehicles and 27,000 trips for weekdays). The cold-soak tank temperature follows the ambient profile with a quarter-hour recursion. During a trip the tank warms as

$$T_{off} = T_{on} + \frac{A + B\,(95 - T_{on})}{1.2}\,\frac{\Delta t}{60}, \qquad T_{off} \le 140\ \mathrm{F}, \tag{5}$$

with A and B by tank temperature group (for passenger cars from model year 1996, $A = 24$ and $B = -0.352$). After the trip the tank cools minute by minute, $T = T_{off} + 1.4\sum(T_{amb} - T)/60$, until it is within 3 F of the cold-soak temperature; the next trip starts from the soak temperature. From this simulation MOVES derives the average tank temperature of each hour for hot soak (mode 150), cold soak (151) and operating (300), the share of parked time in hot and cold soak, and the cold-soak initial-hour fractions (the hour at which each cold-soaking vehicle began to soak). MOVES uses Java `float` in this code, and movesr emulates it.

Tank fuel. `TankFuelGenerator` computes the tank RVP from the market-share average fuel with the Reddy blending relation, weathered with an evaporation temperature derived from the monthly minimum and maximum temperatures, and corrected for commingling of ethanol and non-ethanol fuels.

Calculators. With f_m the operating-mode fractions (hot and cold soak from the soak shares, operating the remainder):

$$\text{permeation} = \mathrm{MBR} \cdot F_{EtOH} \sum_m A\, e^{B\,T_{tank,m}} f_m, \tag{6}$$

$$\text{leaks} = \sum_m \mathrm{MBR}_m f_m, \tag{7}$$

$$\text{venting} = \sum_{bins} s_b\,\mathrm{TVV}_h f_{151} + \mathrm{MBR}_{150} f_{150} + \mathrm{MBR}_{300}\,\tau(t)\,\rho(t,\mathrm{RVP})\,f_{300}, \tag{8}$$

followed by the I/M combination. F_{EtOH} is the permeation fuel adjustment by ethanol level, and τ and ρ are the temperature and RVP adjustments of `EvapTemperatureAdjustment` and `EvapRVPTemperatureAdjustment`. The hourly cold-soak vented vapor TVV_h is the most involved part of MOVES. For each pair of initial hour I and current hour h up to the hour of peak tank temperature, the tank vapor generated is

$$\mathrm{TVG}(I,h) = V_{tank}(1-\phi)\,a\,e^{b\,\mathrm{RVP}}\left(e^{cT_h} - e^{cT_I}\right), \tag{9}$$

interpolated between low- and high-altitude coefficients by barometric pressure and between E0 and E10 by tank ethanol. The canister load over five soak days is

$$X_1 = \mathrm{TVG}_{I..h}, \quad X_2 = (1-\beta)\min(\mathrm{TVG}_{I..24}, C)+\mathrm{TVG}_{1..h}, \quad X_n = (1-\beta)\min(X_{n-1}+\mathrm{TVG}_{h+1..24}, C)+\mathrm{TVG}_{1..h}, \tag{10}$$

with back-purge factor β and canister capacity C. The vented vapor is the regulatory-class weighted combination of the `cumTVVCoeffs` vent and leak equations, which are stored as SQL text; movesr evaluates them in R. Hourly increments are kept only while the tank temperature rises, weighted by the initial-hour fractions (day 1) and by the multiday soaking fractions (days 2 to 5), and the four hours after the peak receive 2, 1, 0.4 and 0.05% of the peak-hour value.

All evaporative rates are computed by regulatory class (MOVES always enables its `WithRegClassID` sections), speciated, and then summed. Permeation bins carry no regulatory class, so they are split with `RegClassSourceTypeFraction`, which movesr rebuilds from `UpdateExecution.sql`. On roads only mode 300 applies and the rate per distance is the operating rate divided by the bin speed.

2.9. Simplified evaporative method

The trip simulation is the slowest step of the evaporative chain. The simplified method keeps everything else and replaces only the simulation. The cold-soak tank temperature is still computed exactly. The rise of hot-soak and operating tank temperatures over the cold-soak temperature, the soak shares and the initial-hour fractions are interpolated from a library of exact results. The library holds 113 MOVES default 24-hour temperature profiles per day type, chosen to cover U.S. climates (20 bins of daily mean temperature, 5 bins of diurnal range, and 13 profiles in the tails). For a new profile the three library profiles nearest in daily mean and diurnal range are combined with inverse squared distance weights, and a warning is issued when the nearest is more than 5 F away. Matching hour by hour on ambient and cold-soak temperature was tested and was not more accurate.

2.10. Comparison statistic and the precision bound

For each matched pair of rates we compute the relative difference $\delta = r_{\mathrm{movesr}}/r_{\mathrm{MOVES}} - 1$, with $\delta = 0$ when both are zero. A pair where MOVES is zero and movesr is not, or the reverse, would give an infinite δ and is counted as a failure; none occurred. MOVES writes its output tables from single-precision columns, which MariaDB prints with six significant digits. Rounding a positive number to six significant digits changes it by at most 5×10^{-6} in relative terms, and single-precision storage adds about 6×10^{-8} per operation. Differences up to about 10^{-5} therefore carry no information about the method. We report the median, the 99th percentile and the maximum of $|\delta|$, and the number of rates above 10^{-5} and 10^{-4}.

2.11. Replication protocol

The steps needed to obtain the same numbers with MOVES, movesr and VEIN are:

1. Use MOVES6.0.0 at commit `45805553` with the default database `movesdb20260729`, and the RunSpecs in the repositories. Any other MOVES version or database changes the reference.
2. Convert the default database to Parquet once and check the stored SHA-256 hashes.
3. Give movesr the same location (county), calendar year, month, day type, hours, hourly temperature and relative humidity, and the same road types and source types as the RunSpec. Evaporative rates need the 24 hours of the day, because tank temperatures depend on the whole diurnal cycle.
4. Run `paper/scripts/01_parity.R`, which recomputes all comparisons from the MOVES exports, then `02_simplified.R`, `03_vein_roundtrip.R`, `05_venting_hours.R`, `06_timing.sh` (which repeats the MOVES runs), `07_moves_repeat.R` and `04_figures.R`. All numbers in this paper are read from the files these scripts write.
5. To use the factors in VEIN, export them with `movesr_export_vein()` and read them with `moves_factors()`, `moves_start_factors()` and `moves_evap_factors()`. These functions keep MOVES units and convert g/mile to g/km only by dividing by 1.609344.

The movesr test suite contains the same checks on a subset of the golden sets and runs them when the MOVES exports are available.

Table 2: Relative difference $|\delta|$ between movesr and MOVES6 for every rate of the reference runs. Missing: rates written by MOVES and not produced by movesr.

Run	Rates	Compared	Missing	Median	99th pct.	Max.	> 1e-5	> 1e-4
Washtenaw	running	86,304	0	7×10^{-7}	4.2×10^{-6}	6.5×10^{-6}	0	0
Washtenaw	start	5,394	0	6.2×10^{-7}	4.7×10^{-6}	7.2×10^{-6}	0	0
Washtenaw	evap parked	3,186	0	2.3×10^{-6}	1.6×10^{-5}	2.9×10^{-5}	104	0
DMV	running	2,522,784	0	7×10^{-7}	4.2×10^{-6}	1.6×10^{-5}	156	0
DMV	brake/tire	529,536	0	6.7×10^{-7}	4.1×10^{-6}	7.8×10^{-6}	0	0
DMV	start	155,112	0	5.7×10^{-7}	4.2×10^{-6}	6.6×10^{-6}	0	0
DMV	evap parked	47,610	0	2.3×10^{-6}	1.1×10^{-5}	2.1×10^{-5}	806	0
DMV	evap driving	761,760	0	2.4×10^{-6}	2×10^{-5}	3.5×10^{-5}	78,062	0
DC 24 h	evap parked	190,440	0	2.4×10^{-6}	2.9×10^{-5}	8.7×10^{-4}	7,232	353

3. Results

3.1. Agreement with MOVES

Table 2 and Figures 2 and 3 summarise the comparison (the Washtenaw evaporative rates are in the table but not in the figures). movesr reproduced all 4,302,126 rates, with no missing or extra rates. For running and start exhaust, brake and tire wear, the largest difference was 1.6×10^{-5} and almost all differences fall in the decades 10^{-8} to 10^{-6}, the range expected from MOVES output rounding. The few exhaust rates between 10^{-5} and 2×10^{-5} are energy consumption (pollutant 91), which MOVES reports in joules with large magnitudes and therefore more single-precision rounding before printing.

Evaporative rates agree within 2.9×10^{-5} at the 99th percentile. The largest differences are in vapor venting of E85 vehicles during the hours of rising temperature. Venting is computed from hourly differences of cumulative single-precision values and from exponentials of tank temperatures, so a change of 10^{-5} F in a tank temperature is amplified. MOVES writes tank temperatures with four decimals, which is not enough to remove this sensitivity: when movesr was fed the cold-soak temperatures as printed by MOVES, instead of computing them, the maximum difference grew to 2.7×10^{-3}. We therefore attribute the residual to single-precision rounding inside MOVES rather than to a different method, although we could not verify this directly because the unrounded MOVES values are not available. On roads, 77,207 of the 253,920 permeation rates differ by more than 10^{-5} (maximum 3.5×10^{-5}). Permeation depends exponentially on the operating tank temperature, which movesr computes in the same single-precision steps as MOVES but cannot match bit for bit.

The Washtenaw evaporative rates are for a weekend day in January and were the only weekend evaporative rates available. All 3,186 of them were reproduced within 2.9×10^{-5}, but only after one setting was changed. With the E85 tank fuel built from the E85 formulation as supplied (the setting that reproduced the DMV events and DC 24-hour runs), 63 E85 venting rates at hour 3 were 1.7% too low. With the E85 formulation carrying E10 properties, as `FuelEffectsGenerator` leaves it, they matched. The same E10 setting was needed for the DMV running run (Section 4.1). movesr also returns rates for model years 1985 to 1999, which this reference run does not contain.

MOVES itself is not exactly repeatable at this level. Running the three DMV and DC RunSpecs a second time (for the timing in Section 3.6) gave the same 4,361,724 non-null rates, of which 15,022 differed from the first run, by up to 1×10^{-5}. Differences of this size between movesr and MOVES are therefore within the run-to-run variation of MOVES.

3.2. Intermediate tables

Table 3 compares intermediate evaporative tables for the District of Columbia with the MOVES execution database. Population, SHO and SHP agree to the precision of the exported values. Average tank temperatures agree within 5×10^{-4} F, which is the last printed digit, soak shares within 5×10^{-7}, and initial-hour fractions within 5×10^{-6}. Figure 4 shows the tank temperatures of passenger cars over the day.

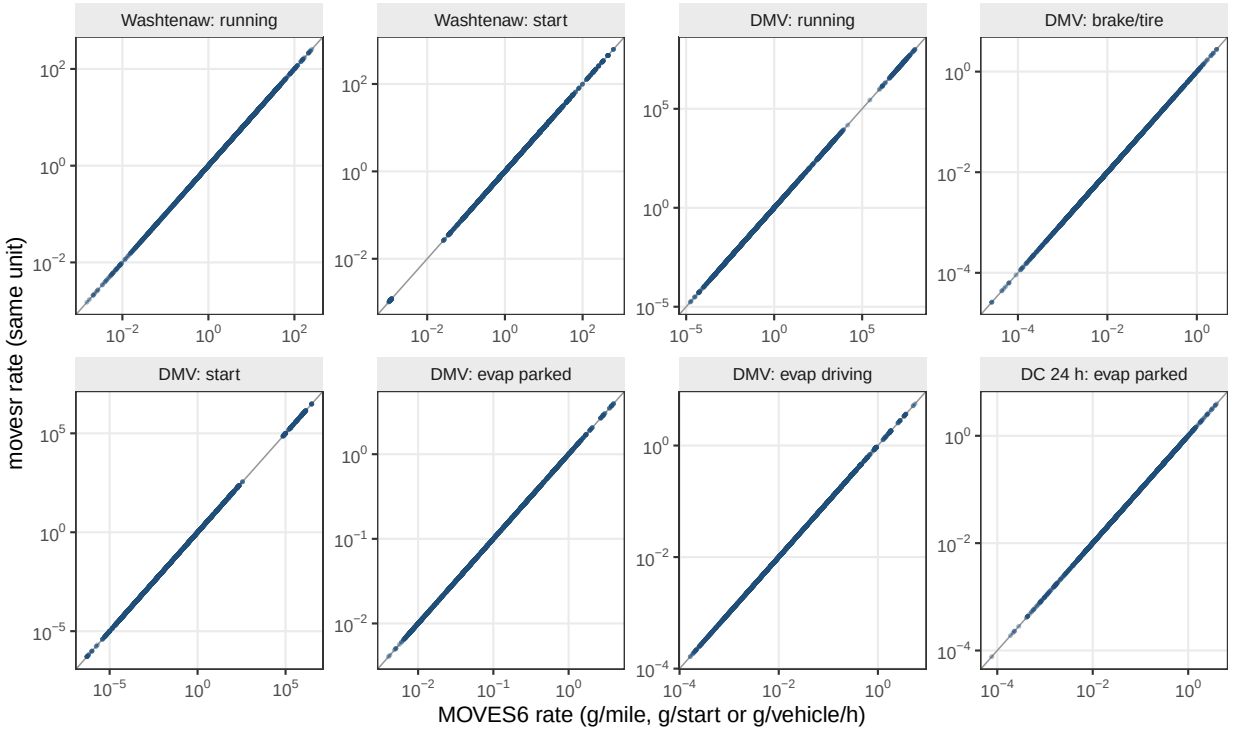


Figure 2: movesr rates against MOVES6 rates for every reference run (random sample of up to 4,000 rates per panel; all rates are compared in Table 2).

Table 3: movesr against MOVES execution tables, District of Columbia, July weekday. Every row exported by MOVES was matched; MOVES exported soak activity fractions only for hours 16 and 17.

Table	Rows compared	Max. difference	Unit
AverageTankTemperature, mode 150	384	5×10^{-4}	F
AverageTankTemperature, mode 151	384	4.9×10^{-5}	F
AverageTankTemperature, mode 300	384	5×10^{-4}	F
ColdSoakTankTemperature	24	4.9×10^{-5}	F
SoakActivityFraction	52	5×10^{-7}	fraction
ColdSoakInitialHourFraction	2,863	3.9×10^{-6}	fraction
SourceTypeAgePopulation	530	4.5×10^{-6}	relative
SHOByAgeRoadwayHour	50,880	3.4×10^{-7}	relative
SHPByAgeHour	12,720	4.6×10^{-7}	relative

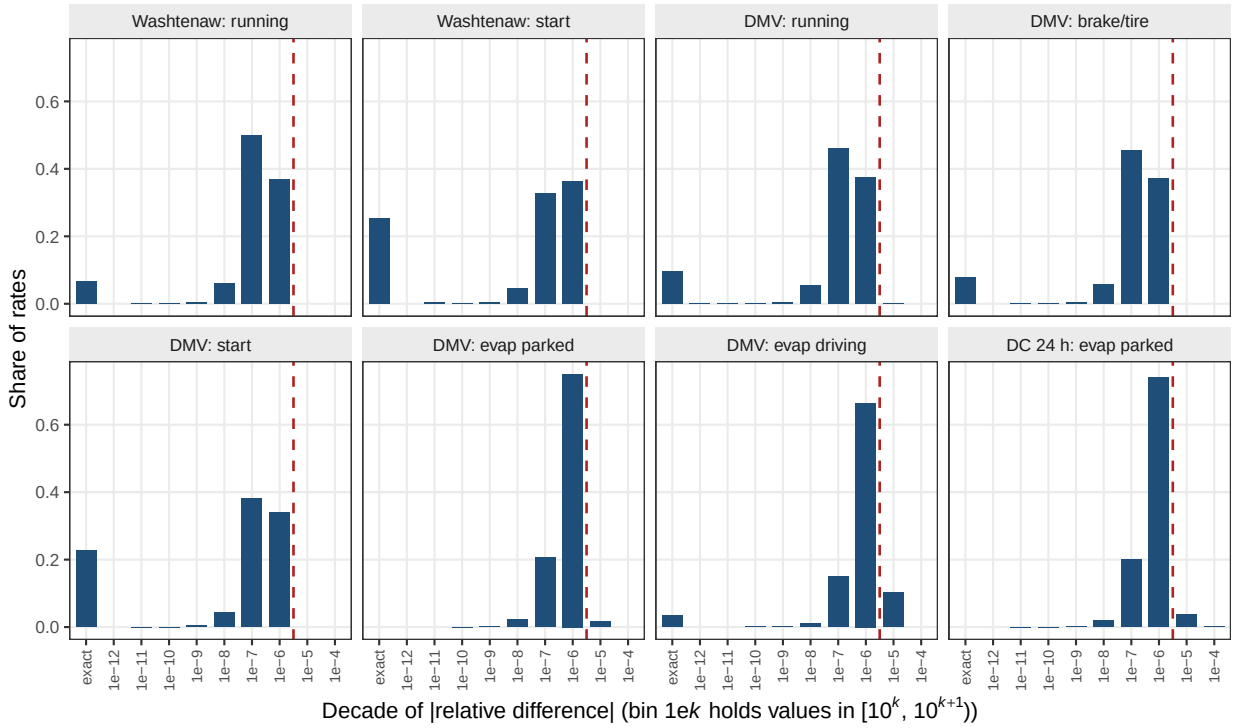


Figure 3: Distribution of the relative difference between movesr and MOVES6 by decade, for all rates. The dashed line marks 10^{-5}; MOVES output rounding alone gives differences up to 5×10^{-6}.

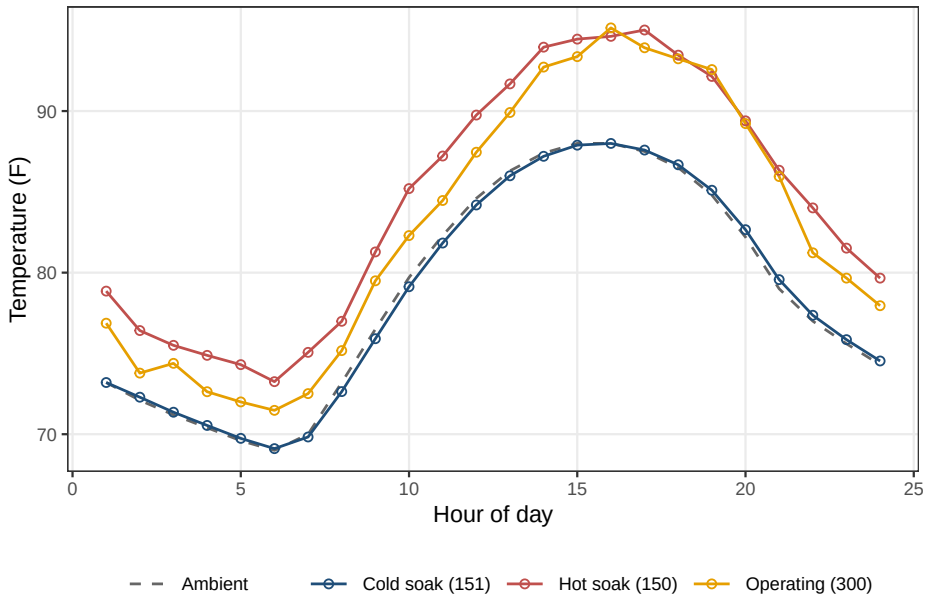


Figure 4: Average tank temperature of passenger cars (tank temperature group 3) in the District of Columbia on a July weekday: movesr (lines) and MOVES execution tables (circles), with the ambient temperature.

3.3. A MOVES behaviour: vapor venting depends on the hours of the run

MOVES restricts the worker's `HourDay` table to the hours selected in the RunSpec, and the vapor-venting script joins both the initial and the current hour of each cold soak to that table. Cold soaks that began outside the selected hours therefore do not vent, and the multiday terms, which are anchored at hour 1, disappear when hour 1 is not selected. In a run with a single hour no cold-soak venting is left at all; only hot-soak and operating venting remain. movesr reproduces this when it is given the same hours (argument `tvv_hours`) and otherwise computes venting over the 24 hours, which is what a full-day MOVES run gives.

Figure 5 shows the size of the effect for gasoline passenger cars in the District of Columbia. With 24 hours, movesr agrees with the MOVES 24-hour run within 3.1×10^{-5}. Computed as single-hour runs, venting between 08 and 15 h falls to between 36 and 98% of the full-day value, depending on hour and model year (for model year 2015 at 10 h, 0.0422 instead of 0.0831 g per vehicle and hour). After the hour of peak tank temperature the two coincide, because cold-soak venting stops when tanks start to cool. The DMV events run used hours 16 and 17; its venting rates are 2.3% lower than those of the 24-hour run at 16 h for model year 2015, and they match the movesr single-hour computation within 1.3×10^{-5}. For inventories, venting factors should come from full-day runs. This matters because evaporative VOC emissions grow quickly with temperature and contribute to ozone formation (Ibarra-Espinosa et al., 2022; Lu et al., 2025; Wu et al., 2024), so a configuration choice that removes most morning venting on a hot day changes the emissions that matter most.

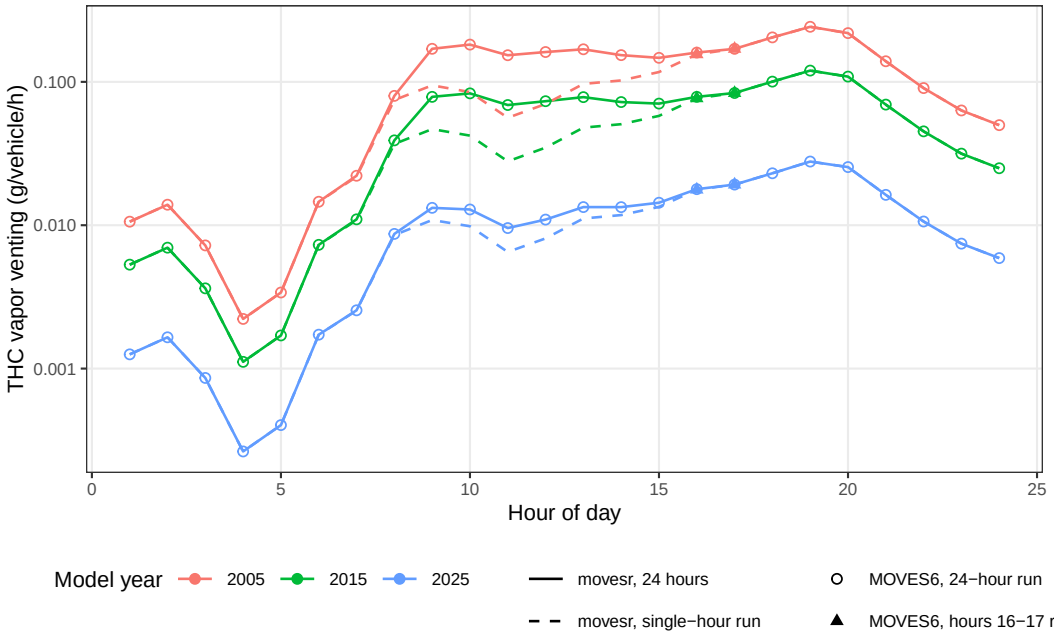


Figure 5: THC vapor venting of gasoline passenger cars in the District of Columbia, July weekday. Solid lines: movesr over 24 hours; dashed lines: movesr computing each hour as a single-hour run, as MOVES does; circles: MOVES6 24-hour run; triangles: MOVES6 run with hours 16 and 17.

3.4. Simplified evaporative method

The simplified method was tested on 20 MOVES default profiles not in the library, with daily means from -14 to 25 C, for weekdays and weekends (Figure 6). Daily totals of permeation and leaks differed from the exact port by at most 0.44%. Vapor venting differed by 3.8% on average. Excluding profiles farther than 5 F from the library, the largest venting difference was 10%. The method issued its warning for 2 test profiles (daily mean -14.3, 21.4 C), where the venting difference reached 54%. In a diagnostic run on two test

profiles, made with an earlier and smaller library, replacing the interpolated hot-soak shares by the exact ones removed almost all of the venting error, while exact tank temperatures or initial-hour fractions changed little. The hot-soak share depends on how fast tanks cool towards the ambient temperature over the whole day, which two daily statistics describe only roughly.

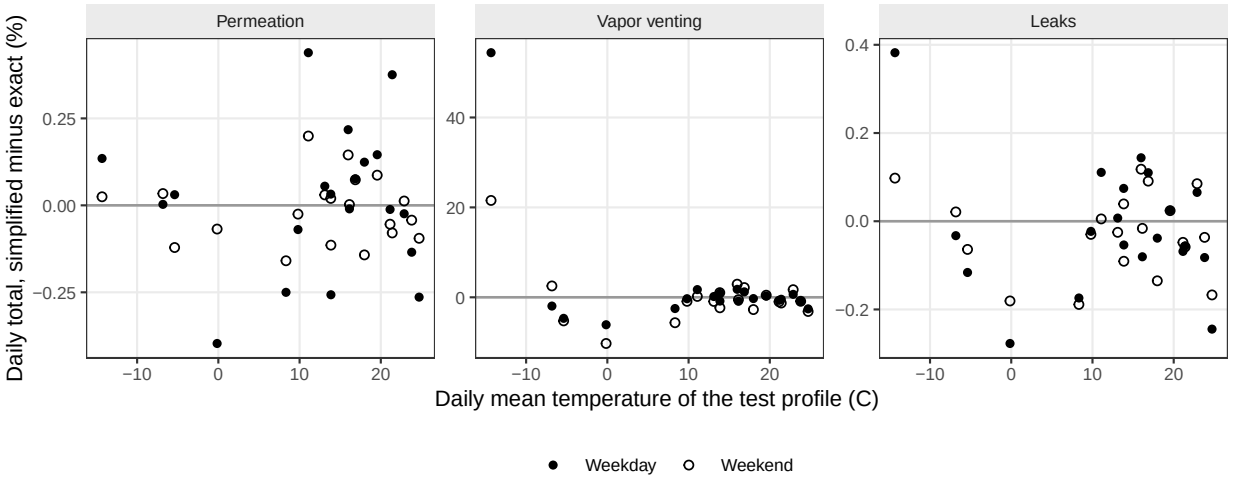


Figure 6: Daily totals of the simplified evaporative method relative to the exact port, for MOVES default temperature profiles not used to build the library.

3.5. From movesr to VEIN in the District of Columbia

Running NOx, CO, VOC and PM2.5 and brake PM2.5 rates for passenger cars, passenger trucks and combination long-haul trucks in the District of Columbia (July, 16 h) were exported with `movesr_export_vein()`, read with `vein::moves_factors()` and evaluated with `vein::ef_moves()` at one speed inside each MOVES speed bin. All 720 values agreed with the MOVES DMV run within 5.1×10^{-6} (Figure 7). For this, VEIN was extended to read brake, tire and on-road evaporative rates, and its documentation of evaporative units was corrected: MOVES writes all three evaporative processes as grams per vehicle in the hour, including vapor venting in `RatePerProfile`.

3.6. Computing time

Table 4 compares the time needed to produce the same rates. The measurements were made one after the other on a shared workstation (Intel Xeon Gold 5215, 40 hardware threads, 85 GB of memory) while an unrelated R process kept about one core busy and held 53 GB of memory, so absolute times will differ on other machines. MOVES ran with one worker, as in all reference runs. MOVES can distribute its work over several workers, which we did not test; movesr runs in one thread. With these settings movesr produced the same rates 3.7 to 9.2 times faster. The largest gain was for the running set (9.2 times). We did not profile MOVES, so we cannot say where its time goes; its log shows that the running run was split into 672 bundles, each sent to the worker, computed with its own SQL scripts and tables, and returned to the master. MOVES-Matrix (Liu et al., 2019, 2017) addresses the same cost differently: it runs MOVES in advance on many computers and looks rates up afterwards, which is fast at the time of use but limited to the stored input grid. movesr computes any combination on demand; the two approaches are complementary, and movesr could also be used to fill such a matrix.

The gains have costs. movesr keeps whole tables in memory and needed 3.1 times more memory than the largest MOVES process for the running set. The simplified evaporative method saves less time than one might expect: 38 s against 53 s for the DC 24-hour workload, because the tank temperature simulation is only part of the work; starting R, reading the database, computing activity and running the calculators take

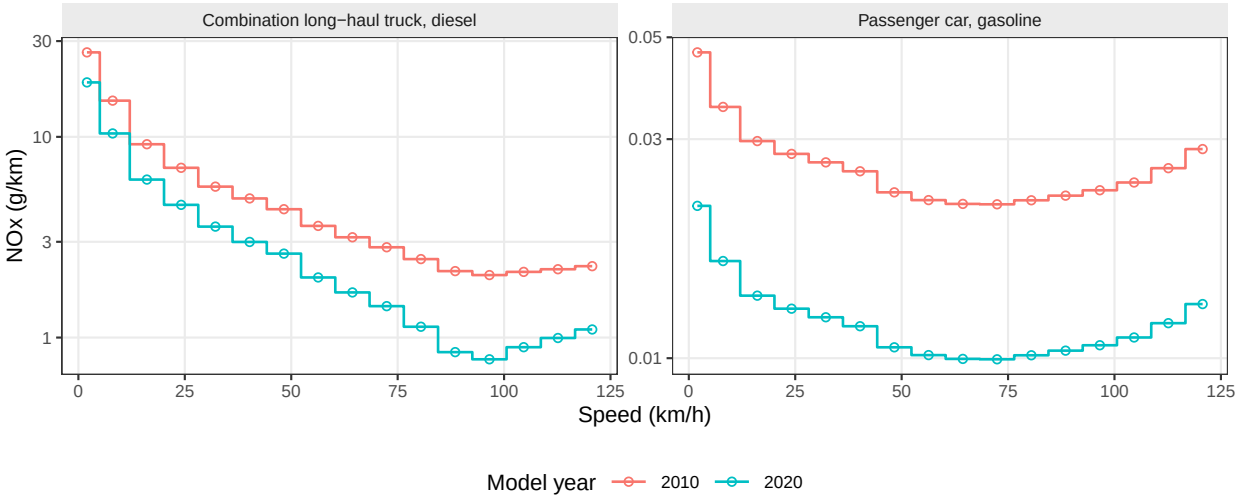


Figure 7: NOx emission factors in the District of Columbia (July, 16 h) evaluated by VEIN from movesr factors (steps) and the MOVES6 rates of the same bins (circles).

Table 4: Wall-clock time (minutes) and peak memory (GB) of the MOVES6 reference runs and of movesr computing the same rates on the same workstation. MOVES6: one run each, including start and shutdown of its MariaDB server; the Java column is the time reported by MOVES itself. movesr: three runs in fresh R processes, including R start-up and reading the database. Ratio: MOVES6 total over movesr median. Memory: largest resident set of a single process (GNU time), so the MOVES value is the larger of its Java program and its database server.

Workload	MOVES6 total	MOVES6 Java	movesr median	movesr range	Ratio	MOVES6 memory	movesr memory
DMV running set: running, brake, tire (21 pollutants) and on-road evaporative (5 HC), 3 counties, 2 h	57.9	55.4	6.3	6.2 to 6.3	9.2	0.8	2.4
DMV events set: start (21 pollutants) and parked evaporative (5 HC), 3 counties, 2 h	9.9	9.3	2.7	2.6 to 2.7	3.7	0.7	0.8
DC evaporative, 24 h, exact tank model	5.9	5.4	0.9	0.9 to 0.9	6.7	0.6	0.7
DC evaporative, 24 h, simplified tank model			0.6	0.6 to 0.6			0.7

the rest. Its advantage appears only when many temperature profiles are processed in one session, where each exact simulation took 5 to 14 s in our tests and each interpolation 0.2 s.

4. Discussion

4.1. What makes the numbers equal

Equality across MOVES, movesr and VEIN depends on a short list of conditions. The MOVES version and default database must be the same, because EPA updates rates and activity between releases. The inputs must be the same, including details that are easy to overlook: the day type, the hours selected (vapor venting), the 24-hour temperature profile (all evaporative processes, even when only one hour is reported), and whether ages with no population are expected to give zero or no rate. Several MOVES steps are only defined by their code, and some of them differ from what the documentation or the SQL suggests. The ones we found while porting are:

- drive-cycle operating modes come from the Go generator, with zero acceleration after missing seconds;
- `EmissionRateAdjustment` and temperature adjustments can have several model-year ranges for the same key;
- energy is written in the unit of the RunSpec (joules here);
- ages with `ageFraction = 0` give a zero rate in `RatePerDistance` and `RatePerStart`, and a NULL rate for on-road evaporative emissions;
- evaporative rates are computed per regulatory class even when the output is not;
- `stmyTVVEquations` stores the leak fraction already multiplied by the regulatory-class fraction, and the vented vapor is multiplied by that fraction again;
- vapor venting depends on the hours selected in the run; single-hour runs lose all cold-soak venting (Figure 5);
- the E85 tank fuel used for vapor venting is not the same in every run. `FuelEffectsGenerator` replaces the properties of E85 formulations with E10 values (cloning formulations per fuel-supply usage), and `TankFuelGenerator` reads the same table. In two of our runs (DMV events and DC 24 h) the venting rates match the E85 formulation as supplied; in the other two that contain venting (DMV running, Washtenaw events) they match the E10-substituted formulation. We did not find what decides this; it is not simply whether exhaust processes are in the run, since the Washtenaw and DMV events runs have the same processes. The effect on E85 venting was 0.4% in the DMV running run and up to 1.7% in the Washtenaw run. movesr offers both (`e85_tank_properties = "raw"` or `"e10"`) and uses the first by default; users who need to match a particular MOVES run may have to try both.

A faithful port has to keep these behaviours, or document clearly where it departs from them. movesr keeps them by default, with two exceptions chosen for inventory use: venting is computed over 24 hours, and the simplified evaporative method is available as an option.

4.2. Is it good enough?

movesr does not need to be identical to MOVES to the last bit to be useful, and it is not. What matters for an inventory is whether a difference could change a result. For exhaust, brake and tire rates the differences are at the precision MOVES itself writes, so an inventory built from either set of factors is the same to six significant digits. For evaporative rates the largest difference, 0.087%, occurs in a few E85 venting rates; a difference of this size does not change an inventory reported to two or three significant digits. The simplified evaporative method is a different matter: errors of a few percent in vapor venting, and much larger ones outside its library, are acceptable for screening or for very large batches of locations but not when MOVES-equivalent results are required. Because the exact tank model now takes seconds per county, month and day type, the exact method is affordable for most inventories, and we recommend it by default.

Two further limits should be stated plainly. First, movesr is not EPA software. It reproduces MOVES for research and inventory work, but regulatory analyses such as State Implementation Plans and transportation

conformity must use MOVES. Second, the code was written by a language model (Section 6). Where movesr was compared with MOVES, that origin does not matter, because the comparison is the evidence. Where it was not compared (next section), errors are possible that a human porter might or might not have made, and users should compare with MOVES for configurations that differ substantially from the reference runs.

4.3. Limits of the verification

The comparison covers what the reference runs contain: two calendar months, one year (2025), weekday rates for the DMV counties and weekend rates for Washtenaw, urban unrestricted roads for distance rates, and low-altitude counties. Weekend evaporative rates were compared for one county and three hours only, and needed the E85 setting discussed above. Other years (movesr uses the default population of the requested year, which approximates MOVES growth for non-base years), rural and restricted-access roads, high-altitude counties, user-supplied fuels and inspection programs, and off-network links with operating time were not compared with MOVES. They use the same code paths, but they have not been tested against the model. Extended idle, auxiliary power units, crankcase processes, refueling and air toxics are not ported.

movesr reproduces MOVES, and MOVES describes the U.S. fleet. Outside the United States, weather, fuel, barometric pressure and inspection programs can be replaced, but the technology of each model year, the fleet composition within source types and the trip patterns remain those of MOVES. Local inspection programs and their high emitters can differ strongly from U.S. assumptions (Ibarra-Espinosa et al., 2025), and local emission factors measured, for example, in tunnels (Nogueira et al., 2021) may be needed to check or adjust MOVES technology assumptions.

4.4. Use for the Washington DC inventory

For the DC application the workflow is: generate running, start, brake, tire and evaporative factors for each county of the DC-MD-VA region with `movesr_region_rates()`, using MOVES default weather or local observations for the 24 hours of each month and day type; export them with `movesr_export_vein()`; and combine them in VEIN with link-level traffic. Because every factor type has been checked against MOVES for three of these counties, differences between the VEIN inventory and a MOVES inventory of the same area should be attributable to activity, not to emission factors.

5. Conclusions

movesr reproduces the MOVES6.0.0 emission rates of running and start exhaust, brake and tire wear and evaporative processes, including their chained pollutants, without Java, MariaDB or MOVES. All 4,302,126 rates of five reference runs were reproduced, within the six-digit precision of MOVES output for exhaust, brake and tire, and within 8.7×10^{-4} for evaporative rates. The factors pass to VEIN unchanged, and producing them took 3.7 to 9.2 times less time than MOVES with one worker on the same machine. A simplified evaporative method gives permeation and leak totals within 0.44% and vapor venting within a few percent on average, but with larger errors for climates outside its library; the exact method remains the default. The scripts that produced every number in this paper are part of the repository, so the comparison can be repeated whenever MOVES or movesr changes.

6. Declaration of generative AI and AI-assisted technologies in the research and writing process

The movesr source code, the comparison scripts, the VEIN changes described above and the first draft of this manuscript were written by Claude Opus 5.5 (Anthropic, 2026), a large language model from Anthropic (model identifier `claude-opus-5.5`), run as a coding agent in OpenCode (OpenCode, 2026) during working sessions on 28 and 29 October 2026. The agent read the MOVES source code, wrote and ran R code, compiled and ran MOVES on the author's workstation to produce the reference runs, compared the outputs and corrected the code when comparisons failed. The author set the objectives and constraints (no Java or MariaDB at run time, data.table, a separate package connected to VEIN through factor files, original and

simplified evaporative methods) and decided the questions the agent raised. The evidence that the code is correct is the comparison with MOVES reported here, which any reader can repeat with the published scripts. After using these tools the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

7. Code and data availability

movesr is available at `https://github.com/atmoschem/movesr` under the GPL (>= 2) licence, like MOVES. The manuscript, scripts and the compact comparison results used for tables and figures are in its `paper` directory. The VEIN interface is available at `https://github.com/atmoschem/vein`. The MOVES exports used as reference can be regenerated with the RunSpecs and the script in `data-raw/dmv`; the Washtenaw reference set is distributed with VEIN. The manuscript was prepared with rticles (Allaire et al., 2024) and the figures with ggplot2 (Wickham, 2016).

8. CRediT authorship contribution statement

Sergio Ibarra-Espinosa: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing - original draft, Writing - review and editing, Project administration. The use of a language model for software and drafting is described in Section 6.

9. Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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